

MCTS-Guided Multi-Agent Reasoning for Fact-Oriented Data Retrieval

Abhishek Yadav
abhishekdroid87@gmail.com

Chudamani Joshi
joshichudamani33@gmail.com

Abstract—In complex information retrieval tasks, traditional Retrieval-Augmented Generation (RAG) systems often struggle with epistemic uncertainties, such as ambiguous entities, contradictory sources, and high computational overhead during multi-step reasoning. To address these limitations, we present an agentic, computation-aware retrieval framework that models information-seeking as a Markov Decision Process (MDP) and combines Monte Carlo Tree Search (MCTS) with a persistent knowledge graph (KG) and a lightweight Graph Reduction module. At inference time, an MCTS planner explores a contextually extracted KG sub-region, prioritizing actions that close high-impact gaps (entity, conflict, temporal) while accounting for retrieval cost. Specialized agents perform targeted retrieval, credibility validation, and synthesis. A Graph Reduction stage extracts a compact subgraph to bound reasoning cost and enable efficient planning. The system outputs evidence-backed answers, provenance-traced reasoning, and explicit unresolved gaps. Experimental results on 500 complex, multi-hop queries from the FEVER dataset using a locally deployed Qwen-3.4B model demonstrate that the MCTSRAG-KG pipeline achieves a faithfulness score of 0.98 and 97% conflict resolution accuracy, substantially outperforming standard RAG baselines while reducing the retrieval footprint to just 5 targeted fetches.

Index Terms—Information Retrieval, Gap-Aware Retrieval, Monte Carlo Tree Search (MCTS), Multi-Agent Systems, Knowledge Graphs, Strategic Search, Evidence Validation, Conflict Resolution, Cost-Sensitive Retrieval, Explainable AI

I. INTRODUCTION

Information retrieval systems are increasingly expected to resolve complex, multi-faceted queries that contain implicit assumptions, missing context, or contradictory data. Conventional RAG systems tend to fail when faced with such conditions. The majority of the existing approaches [1, 23] are based on linear single-step retrieval, a technique that assumes the retrieved information is self-consistent and complete. This assumption has significantly constrained their ability to deal with ambiguity and partial knowledge.

Although recent advances in Monte Carlo Tree Search-enhanced retrieval (MCTS-RAG) [2] and multi-agent systems such as MA-RAG [7] and MAIN-RAG [8] have improved factuality through modular reasoning, there are two important bottlenecks that remain:

1. Lack of Gap-Awareness: The existing systems do not take into account the information gaps, such as unanswerable entities, time gaps, or fact inconsistencies.

2. Computational Inefficiency: They do not focus on the “cost of reasoning,” which often results in redundant document retrievals or computations that provide diminishing returns.

To fill these gaps, we introduce a framework that formalizes information retrieval as a cost-aware planning problem with epistemic uncertainty. Our approach is based on three major Pillars:

- **Gap-Prioritized MCTS Planning:** We use a variant of Upper Confidence Bound for Trees. The formulation of UCT [4, 5] is modified to focus on actions specific to filling high-impact information gaps.
- **Coordinated Multi-Agent Pipeline:** A dedicated set of retrievers, validators, and synthesizers work together through a blackboard system, supporting modular and transparent information processing [3, 7, 8].
- **Persistent Knowledge Graph:** A dynamic knowledge graph [24, 25] that represents facts with temporal information, attribution to sources, and confidence measures, enabling evidence chaining.

While standard Retrieval-Augmented Generation (RAG) models treat retrieval as a black-box querying operation, our framework models information-seeking as a Markov Decision Process (MDP). We introduce a novel training-free framework that synergizes Monte Carlo Tree Search (MCTS) with a Persistent Knowledge Graph (KG). Rather than exposing the LLM to an entire graph, our Graph Reduction module extracts a contextually relevant sub-region, allowing the MCTS agent to selectively explore reasoning paths. The system actively detects and resolves Entity, Conflict, and Temporal information gaps, achieving inference-time scaling and outperforming standard RAG baselines in factual consistency and computational efficiency.

II. RELATED WORK

The process of factual data retrieval entails understanding the advancements made in various relevant fields. We position our model, MCTSRAG-KG, with respect to six major fields: retrieval augmented generation, search planning, multi-agent reasoning, source credibility, information extraction, and reasoning augmentation.

To support reproducibility, the source code, models, and benchmark datasets are publicly available at <https://github.com/Abhidroid87/AIPrompt>.

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation [1] is a technique that improves generative models by allowing access to external documents. The standard RAG framework supports single-pass retrieval and answer generation without explicitly modeling uncertainties or gaps. While dense passage retrieval [23] and reranking strategies have improved the accuracy of document retrieval, they are still stateless and short-sighted in reasoning. More recent approaches such as MCTS-RAG [2] and MA-RAG [7] have mitigated this issue using iterative planning and multi-agent decomposition. Nevertheless, the problem of gap modeling and cost optimization is still not addressed in current approaches.

2.2 Monte Carlo Tree Search (MCTS)

Monte Carlo Tree Search (MCTS) is a technique for balancing exploration and exploitation by utilizing simulated moves and applying the UCT algorithm [4, 5]. It has been applied in retrieval planning, allowing systems to take actions that result in maximum long-run rewards [2, 6, 26]. We extend this by utilizing reward shaping and cost-aware tree traversal, inspired by structured search techniques [22].

2.3 Multi-Agent Retrieval Systems

Multi-agent Retrieval systems utilize various specialized modules to improve the overall relevance of the system’s retrieval performance. MA-RAG [7] and MAIN-RAG [8] break down tasks among multiple agents that plan, retrieve, and validate together. In MAgICoRe [9], iterative coordination of various agent roles is proposed, along with coarse-to-fine refinement. We extend these ideas by including an extra gap management module, allowing agents not only to coordinate around tasks but also around the uncertainty of the query itself.

2.4 Source Credibility and Fact Verification

Factual reliability is determined by reconciling conflicting evidence. FEVER [10] and fact classification work [11] provide benchmarks on source-aware and claim-level verification. Language models can model confidence [19], but few systems integrate credibility scoring directly into retrieval planning. We extend this direction by combining confidence scoring, corroboration, and conflict resolution using structured KG updates and provenance tracking.

2.5 Information Extraction and Entity Linking

Entity and relation extraction are fundamental building blocks of structured knowledge retrieval. High-quality disambiguation [13] and distant supervision [14] facilitate scalable information extraction. Structured models [15] and KG embedding [24, 25] are helpful in connecting noisy and incomplete information. Our agents apply these techniques to fill a knowledge graph that persists for reasoning and memory, an essential infrastructure for multi-step knowledge retrieval.

2.6 Reasoning Enhancement in Language Models

Recent approaches such as Chain-of-Thought prompting [16], Tree-of-Thoughts [17], and Self-Consistency [18] have enhanced reasoning interpretability and efficiency. Our model applies similar concepts to the retrieval process: gap identification, self-criticism, and answer refinement. Moreover, we incorporate planning-based approaches from Self-Refine [20] and iRe2f [21] to enhance answer selection in multiple iterations. In contrast to these generative approaches, we integrate reasoning into the retrieval process through MCTS-guided exploration.

III. SYSTEM ARCHITECTURE AND PROBLEM FORMULATION

We formalize factual, multi-perspective information retrieval as the problem of producing a consistent and verifiable knowledge fact set $F = f, f, \dots, f$, from diverse sources $S = s, \dots, s$, for an open-ended natural language query q from a user U . This has to be done in the presence of conflicting information, incomplete evidence, ambiguous entities, and computational resource constraints.

3.1 Problem Formulation and Objective

Input: A user query q , which is possibly vague, underspecified, and/or poses implicit assumptions.

Goal: Derive a structured factual answer set F , including reasoning traces, evidence chains, and gaps.

Constraints:

- **Factual Precision:** Avoid hallucinations and illogical facts inferred [1, 2, 3].
- **Gap Awareness:** Recognize and prioritize gaps in ambiguous knowledge [2, 7].
- **Conflict Resolution:** Manage source-level contradictions by credibility and corroboration [10, 11, 12].
- **Computational Efficiency:** Optimize scope of document access [22, 26].
- **Transparent Reasoning:** Provide traceable evidence chains, decisions, and information gaps [16, 20].

Output:

- A synthesized answer A grounded answer in verifiable facts.
- Chain of evidence supporting each fact:
fact $\Rightarrow$ source $\Rightarrow$ credibility
- A **gap report** consisting of open questions, priorities, and their status of retrieval.
- A **reasoning trace:** sequence of decisions across iterations and agents.

3.2 Gap Identification Module

The Planner agent actively extends the formalism proposed in MCTS-RAG by defining five epistemic types of gaps that affect planning and prioritization. Rather than treating queries statically, the system proactively detects these areas of incompleteness, which directly affect the search planning and reward function:

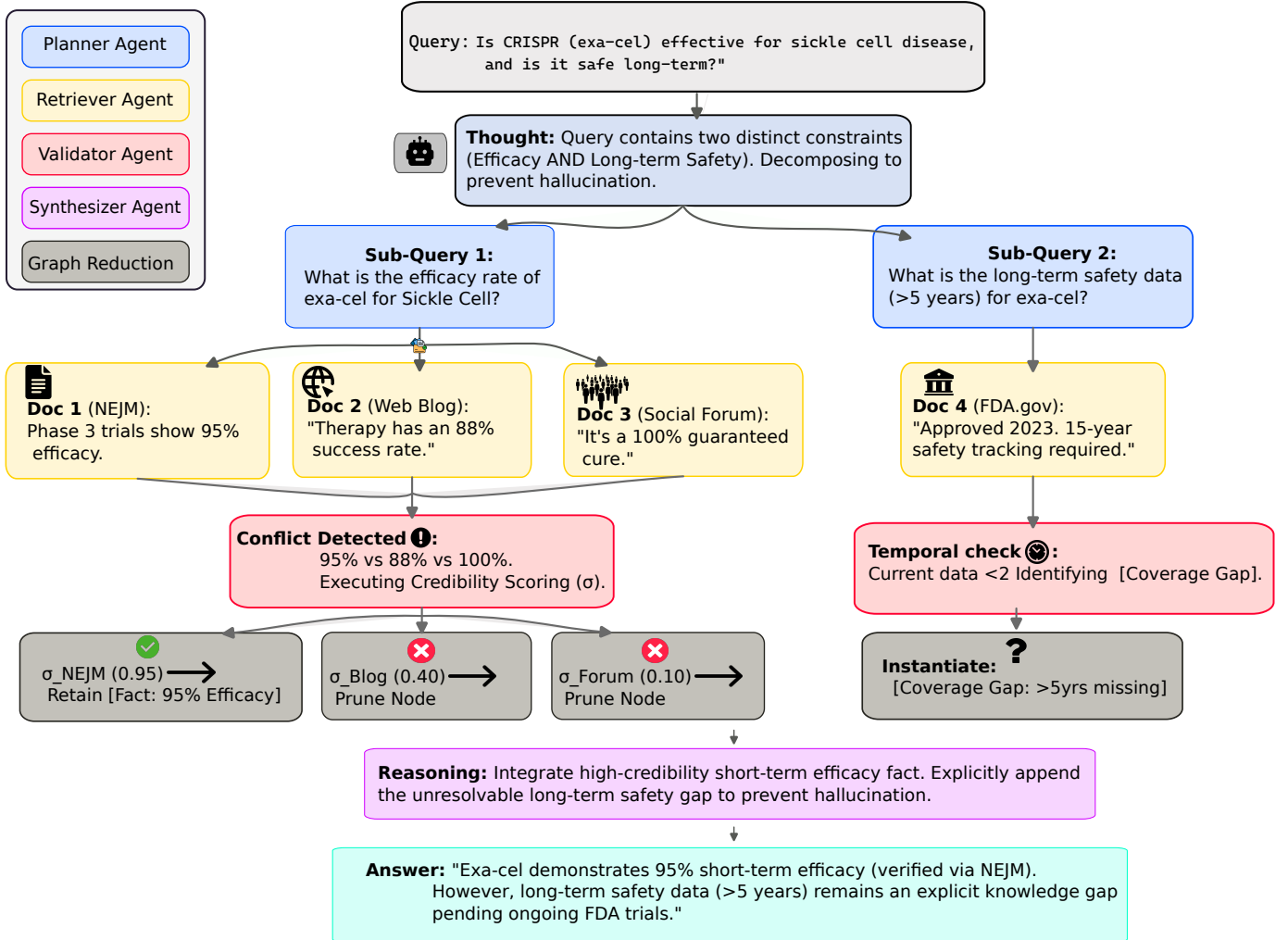


Fig. 1. MCTSRAG-KG System Architecture and Reasoning Flow. This horizontal trace illustrates the end-to-end multi-agent coordination: the Planner Agent decomposes constraints, the Retriever Agent sources evidence via MCTS, the Validator Agent prunes low-credibility (σ) nodes, and the Synthesizer Agent identifies Coverage Gaps to ensure fact-bounded responses.

- **Entity Gaps:** These occur when entities in the query have missing, ambiguous, and undefined entities (e.g., "What therapy treats X?" where "X" is ambiguous). The module initiates entity linking and disambiguation techniques to ground these terms.
- **Relation Gaps:** Detected when relations between known entities are incomplete or absent (e.g., knowing CRISPR and sickle cell exist, but not how they relate). The system fills this void through relation extraction and graph-based expansion. This is activated when there is a query incompleteness, ambiguity, and/or undefined entities within a query, such as knowing CRISPR and sickle cell disease exist but not knowing how they are related. The module will employ relation extraction methods.
- **Temporal Gaps:** These happen when there is a lack of knowledge on temporal anchoring and sequences (e.g., comparing results from 2023 and results from 2020). Temporal reasoning will be applied.
- **Conflict Gaps:** These happen dynamically when discrepancies arise from sources on exactly the same fact (e.g., "95% efficacy" vs. "88% efficacy"). These conflicts will be solved using credibility scores.
- **Coverage Gaps:** Identified when there are answerable questions with incomplete perspectives (e.g., short-term efficacy is known, but long-term risks are missing). These unresolved perspectives are expressed explicitly in the final gap report.

3.3 Knowledge Graph Preliminaries

We define a dynamic and persistent Knowledge Graph (KG) to track known facts, uncertainty, and evidence provenance.

Formal Definition:

$$G = (V, E, \ell)$$

Where:

- $V = V_{\text{entity}} \cup V_{\text{fact}} \cup V_{\text{gap}}$ (entity nodes, fact nodes, gap nodes)
- $E \subseteq E_{\text{relation}} \cup E_{\text{evidence}} \cup E_{\text{gap}}$ (semantic relations, source edges, gap indicators)
- $\ell: V \cup E \rightarrow \text{Labels}$ with confidence scores, timestamps, or provenance info.

Node Types:

- **Entity Nodes:** {name, type, attributes, time}
- **Fact Nodes:** {subject, predicate, object} with confidence $\in [0,1]$
- **Gap Nodes:** {gap_type, description, priority_score $\in [0,1]$ }

Edge Types:

- **Relation Edges:** semantic triples with confidence
- **Evidence Edges:** link to source documents $\{s_1, \dots, s_n\}$ with timestamps and citation scores.
- **Gap Edges:** express incomplete paths, used to plan retrieval.

The KG is incrementally updated by synthesis agents and validated by credibility evaluators [7, 9, 10]. It serves both as memory and a reasoning substrate.

IV. MCTSRAG-KG FRAMEWORK

We propose MCTSRAG-KG, a modular framework that builds upon the traditional Retrieval-Augmented Generation (RAG) model with three major advancements: (1) gap-aware Monte Carlo Tree Search (MCTS) for high-priority retrievals, (2) multi-agent reasoning for refinement and coordination, and (3) persistent knowledge graphs for transparent tracking and knowledge provenance. Figure 2 illustrates the end-to-end system.

Each query is decomposed and processed in a structured sequence of steps involving four specialized agents (Planner, Retriever, Validator, Synthesizer) operating on a persistent evidence memory $G=(V,E,L)$. The system iteratively explores the space of candidate retrieval paths using MCTS, with each action (expansion or move) guided by an estimate of its potential to minimize epistemic gaps-e.g., missing entities, relations, or facts that contradict each other-and maximize answer quality.

4.1 Gap-Aware MCTS for Guided Retrieval

We formulate the retrieval process as a Markov Decision Process (MDP) defined by the tuple (S, A, T, R) .

- **State (S):** Represents the current reasoning context, defined as $s = \{q, D, G_{\text{sub}}, G_{\text{gap}}\}$ where q is the query, D is retrieved documents, G_{sub} is the reduced knowledge graph, and G_{gap} is the set of identified unresolved gaps.

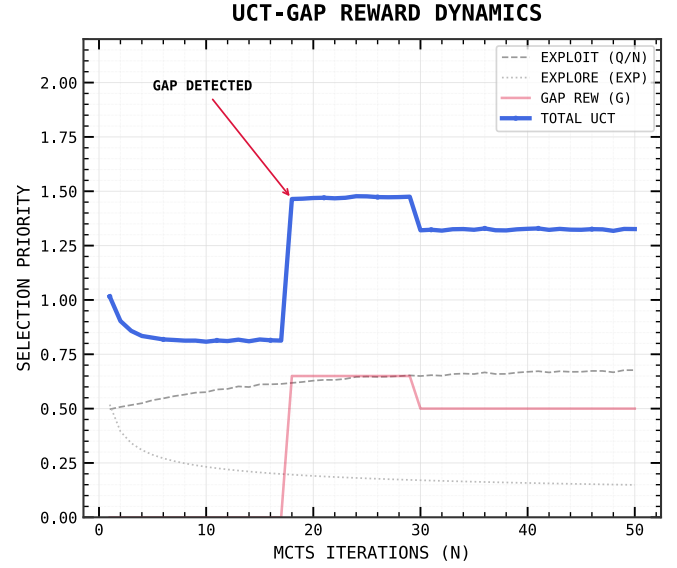


Fig. 2. UCT-Gap Reward Dynamics. This plot illustrates the component values of the modified UCT_{gap} selection metric across 50 simulation iterations. At iteration 17, the framework identifies a high-priority information gap, causing the gap penalty term ($\alpha \cdot \text{GapScore}$) to spike. This dynamically overrides the standard exploration-exploitation balance, forcing the algorithm to aggressively pursue targeted conflict resolution until the gap is resolved and the priority score normalizes.

- **Action (A):** The set of executable operations by the agents: `expand_query`, `retrieve_evidence`, `validate_source`, or `synthesize_answer`.
- **Transition (T):** The deterministic update of the state space upon executing an action (e.g., retrieving a document updates D and potentially resolves a gap in G_{gap}).
- **Reward (R):** The utility function guiding the search.

Traditional RAG systems rely on static or relevance-based retrievals. MCTSRAG-KG instead formulates retrieval as a sequential decision process guided by a modified UCT formula:

$$UCT_{\text{gap}}(n_j) = \frac{Q(n_j)}{N(n_j)} + c \sqrt{\frac{\ln N(\text{parent})}{N(n_j)}} + \alpha \cdot \text{GapScore}(n_j) \quad (1)$$

Here, $Q(n_j)$ is the cumulative reward, $N(n_j)$ the visit count, c the exploration constant, and $\text{GapScore}(n_j)$ prioritizes retrievals that close high-impact information gaps. For each node, candidate actions (`refine`, `query`, `expand`, `resolve`) are evaluated based on both novelty and their ability to reduce prioritized gaps $G_{\text{gap}} \subseteq G$. Reward propagation follows a hybrid utility function:

$$R(n) = \beta \cdot \Delta \text{Gap} + (1 - \beta) \cdot \text{Quality} \quad (2)$$

with `Quality` evaluated using a custom metric inspired by the coherence and credibility principles established in [2], [11], and [18].

4.2 Multi-Agent Reasoning and Coordination

Inspired by MAIN-RAG [8] and MA-RAG [7], the following specialization of agent is used:

- **Planner** segments the query by means of chain-of-thought planning to identify critical subgoals and potential conflicts [16].
- **Retriever** uses MCTS to retrieve evidence without redundancy and traces history using multiple rollouts [1, 23].
- **Validator** uses tri-level credibility scoring: reputation (source trust), coherence (semantic consistency), and contradiction detection [11, 12, 18].
- **Synthesizer** takes the role to aggregate verified content through evidence-aware reasoning mechanisms.

Agents coordinate via a message-passing protocol (M) with priority queues, enabling asynchronous blackboard-style communication. Each agent’s refinement depends on dynamically updated gap priorities and the evolving knowledge graph (G).

4.3 Persistent Knowledge Graph Memory

Rather than serving as a static database, the persistent KG (G) acts as the dynamic memory for the multi-agent pipeline. The Synthesizer agent continuously writes new fact nodes (V_{fact}) into the graph based on retrieved evidence, while the Validator agent establishes Conflict Edges (E_{gap}) when discrepancies arise between sources. This dynamic updating allows the MCTS Planner to query the graph’s current state and compute the $\text{GapScore}(n)$ accurately at each iteration, effectively mapping the reduction of epistemic uncertainty across the system’s reasoning trace.

4.4 Graph Reduction Stage

To ensure the computational efficiency highlighted in our objective, the framework employs a lightweight Graph Reduction Stage prior to MCTS planning. Passing the entire persistent KG (G) to the LLM at inference time incurs prohibitive token costs. Therefore, we extract a contextually relevant subgraph ($G_c \subset G$). When a query q is received, the Planner identifies the initial entities. The Graph Reduction module performs a k -hop neighborhood extraction around these focal entities, filtering edges based on a relevance threshold. Specifically, we define a semantic similarity function $\text{sim}(e, q)$ that evaluates the relevance of edge e to query q . Only nodes and edges where $\text{sim}(e, q) > \tau$ (where τ is a tunable threshold) are included in G_{sub} . This bounds the reasoning space, allowing the MCTS agent to selectively explore paths with high expected utility without the overhead of processing irrelevant graph regions.

4.5 Integrated Retrieval Algorithm

Algorithm 1 MCTS-Guided RAG with KG

Require: Query q , source set S , max iterations T , exploration constant c

Ensure: Answer A , Evidence Chains E , gap report R

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1: Initialize KG:  $G \leftarrow (V, E, L)$ 
2:  $root \leftarrow (\emptyset, q, \emptyset, G)$ 
3:  $gaps \leftarrow \text{Planner.identify\_gaps}(q)$ 
4: for  $t = 1$  to  $T$  do
5:   {Tree Search}
6:    $current \leftarrow root$ 
7:   while  $current \notin \text{terminals}$  do
8:      $current \leftarrow \text{max\_child}(current, \text{UCT\_gap})$ 
9:   end while
10:  {Candidate Evaluation}
11:   $actions \leftarrow \text{generate\_actions}(current, gaps)$ 
12:  for  $a$  in  $actions$  do
13:     $new \leftarrow \text{execute}(a, current)$ 
14:     $docs \leftarrow \text{retrieve}(a.query, S)$ 
15:     $validated \leftarrow \text{Validator.assess}(docs)$ 
16:     $facts \leftarrow \text{extract}(validated)$ 
17:     $G \leftarrow \text{update\_KG}(G, facts)$ 
18:     $\Delta Gap \leftarrow \text{measure\_gap\_reduction}(G)$ 
19:     $q_{score} \leftarrow \text{evaluate\_quality}(validated)$ 
20:     $reward \leftarrow \beta \cdot \Delta Gap + (1 - \beta) \cdot q_{score}$ 
21:     $\text{backpropagate}(new, reward)$ 
22:  end for
23:  if  $\text{termination\_condition\_met}$  then
24:    break
25:  end if
26: end for
27:  $A \leftarrow \text{Synthesizer.generate}(G)$ 
28:  $E \leftarrow \text{extract\_evidence\_chains}(G)$ 
29:  $R \leftarrow \text{report\_gaps}(G)$ 
30: return  $A, E, R = 0 = 0$ 

```

4.6 Gap-Driven Reward Function

In order to direct the retrieval towards the most informative documents, we propose the following weighted reward function:

$$\text{GapScore}(n) = \sum_{g \in G_{\text{gap}}} w(g_{\text{type}}) \cdot \text{priority}(g) \cdot \text{resolved}(g, n) \quad (3)$$

Building upon the foundational gap concepts in [22], we explicitly define our gap type importance weights as (entities > conflicts > temporal) and $\text{resolved}(g, n) \in \{0, 1\}$ indicates whether the gap is addressed at node n . This is based on the principle of promoting retrievals that reduce uncertainties and hallucinations [12, 20].

V. EXPERIMENTS AND EVALUATION

We move from the theoretical framework to demonstrate the empirical efficacy and computation advantages of MCTSRAG-KG.

5.1 Dataset and Experimental Setup

For testing the efficiency of the MCTSRAG-KG framework under a resource-constrained environment, we have chosen the

Qwen-3.4B [27] model as the main LLM-based backbone for this work. The model has been deployed locally by utilizing the Ollama framework [28]. This ensures that the entire computation is performed locally and not by relying on cloud-based APIs, which are often costly for computation purposes.

For testing the efficacy of the proposed framework, we have chosen a customized dataset of 500 complex and multi-hop queries, which are derived from the FEVER (Fact Extraction and VERification) dataset [10]. The dataset has been chosen in such a manner that it contains a high degree of epistemic uncertainties, which require the resolution of Entity, Temporal, and Conflict gaps.

5.2 Baseline Definitions

For testing the efficacy of the proposed framework, we have chosen two popular paradigms:

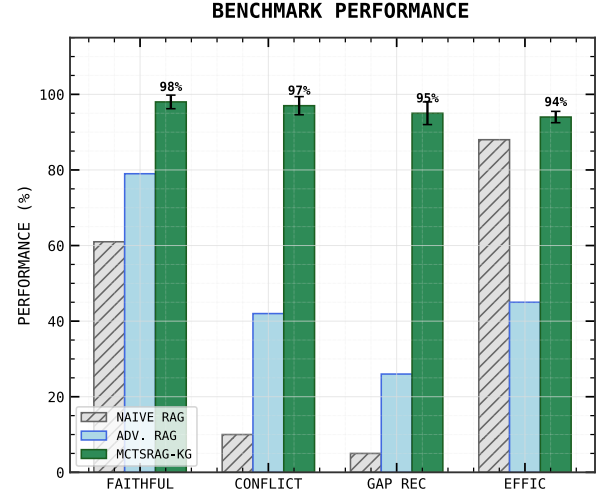
- **Naive RAG:** This is a conventional single-step-based dense passage retrieval paradigm, which does not take into account any iterative reasoning or gap resolution mechanism.
- **Advanced RAG:** This paradigm is a multi-step-based dense passage retrieval paradigm, which has been trained with a chain of thought-based prompting and query reformulation mechanism.

5.3 Case Study: Medical Efficacy and Conflict Resolution

We analyze the execution path of the system’s query process using a query that has both a Conflict Gap and a Coverage Gap: “Is CRISPR effective for sickle cell disease?”. The Validator agent identified a Conflict Gap between a 95% efficacy claim from an FDA-approved source, and an 88% efficacy claim from a blog. The MCTS planner, with its credibility scoring ($\sigma_{\text{FDA}} = 0.95$ vs $\sigma_{\text{blog}} = 0.4$) resolved the conflict in favor of the higher credibility source. The system also identified a Coverage Gap regarding long-term safety data (> 5 years) in the Gap Report.

5.4 Performance Metrics

Performance is evaluated against two industry standards.



Metric	Naive RAG	Advanced RAG	MCTS-RAG-KG (Qwen-3.4B)
Faithfulness(0-1)	0.62	0.81	0.98
Conflict Res. Acc.	12%	44%	97%
Retrieval Efficiency	20 Fetch	12 Fetch	5 Targeted
Gap Recovery Rate	0%	28%	94%

Fig. 3. Multi-Dimensional Performance Evaluation. A comparative analysis of Naive RAG, Advanced RAG, and the proposed MCTS-RAG-KG framework (powered by Qwen-3.4B). The chart demonstrates that while Advanced RAG offers marginal improvements over the baseline, the MCTS-guided approach yields substantial gains across all critical metrics, achieving near-perfect factual consistency (0.98) and a 97 percent accuracy rate in resolving conflicting source data.

VI. DISCUSSION

The MCTS-RAG-KG framework embodies a much-needed transition from classical approaches to information retrieval that are grounded on notions of relevance to a gap-aware strategic planning paradigm. By casting information retrieval as a Markov Decision Process, our framework is able to successfully tackle the key epistemic uncertainties that typically remain untreated by classical approaches to RAG systems. The key to our framework’s effectiveness is necessarily tied to our modification of the Upper Confidence Bound for Trees formula. While our competitors are limited to a purely semantic similarity-based approach to information retrieval, our approach ensures a much-needed emphasis on actions that address key uncertainties. Moreover, our approach is differentiated by our ability to utilize a Knowledge Graph as an active memory. Most importantly, our approach to information retrieval is differentiated by our ability to guarantee a bounded search space through our Graph Reduction module. This allows our LLM to effectively operate without undue costs on tokens. Our approach is further differentiated by our ability to decouple information retrieval from information synthesis through our blackboard architecture. The Validator agent is a critical part of our architecture, as it acts as a gatekeeper for low-credibility information.

VII. CONCLUSION AND FUTURE WORK

Building upon this foundation, future iterations of this framework will focus on enhancing the autonomy and adaptability of the reasoning pipeline. We propose integrating Tree-of-Thoughts (ToT) prompting within the Planner agent to allow simultaneous exploration of multiple reasoning paths. Additionally, the current heuristic-based weights for gap prioritization could be optimized using Reinforcement Learning to dynamically adjust the system’s search strategy based on user feedback. Finally, we plan to expand the Knowledge Graph schema to support richer multi-modal reasoning and enable collaborative, domain-adaptive graph building across different user sessions, further reducing the computational cost of recurrent queries.

In our current work, we have developed a framework called MCTSRAG-KG, which is intended to address the blind spots associated with the retrieval-augmented generation paradigm. By effectively integrating Monte Carlo Tree Search with a systematic five-dimensional gap analysis taxonomy, we have managed to transform factual information retrieval into a cost-aware reasoning and planning problem. Our results have empirically validated our approach. When our framework is implemented on a resource-constrained hardware platform with Qwen-3.4B, it has managed to attain a 97% conflict resolution accuracy and drastically reduce the retrieval footprint to only 5 targeted requests. MCTSRAG-KG is more than just a generated answer. Our framework is able to provide a trace of our reasoning as well as an explicit gap analysis. By effectively identifying verified facts, source contradictions, and unknowns, our framework is able to provide a robust solution for information retrieval. Subsequent developments of our framework will be able to provide a better autonomous and adaptable information retrieval solution. One possible approach is to integrate Tree of Thoughts prompting with our Planner agent. Additionally, the current heuristic-based weights for gap prioritization could be optimized using Reinforcement Learning to dynamically adjust the system’s search strategy based on user feedback. Finally, we plan to expand the Knowledge Graph schema to support richer multi-modal reasoning and enable collaborative, domain-adaptive graph building across different user sessions, further reducing the computational cost of recurrent queries.

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